Cutting Assistant

AI-assisted cutting edge optimization

Cutting Assistant

AI-based cutting-edge optimization for your 2D laser cutting machine

TRUMPF provides proven cutting parameters with its machines in the form of laser technology tables. For differing material grades, however, parameter adjustments are still necessary. With the Cutting Assistant, you can quickly and easily optimize your cutting parameters. Choose between evaluating test parts in bandwidth mode or using AI-generated optimization suggestions with a handheld scanner. The function provides objective recommendations for parameter adjustments, saves time and material costs, and ensures excellent cutting quality — even with varying material grades.

The advantages at a glance

Cutting edge optimization – in your material, for your employees

Dialog-guided cutting data optimization

Even inexperienced users can optimize cutting data like a pro in just a few minutes – saving both time and material costs.

Excellent cutting quality even in non-laser-grade-material

The Cutting Assistant suggests suitable optimizations – whether for special materials or materials with fluctuating quality.

Quick inspection of cutting-edge quality

The handheld scanner allows for an objective assessment of part quality. The Cutting Assistant displays roughness and burr height in micrometers.

Want to see how edge optimization works with the Cutting Assistant?
Watch our product video!
https://www.trumpf.info/fsbpmj

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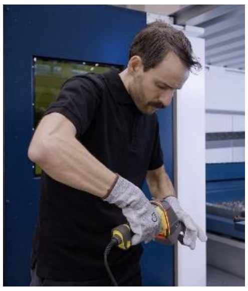
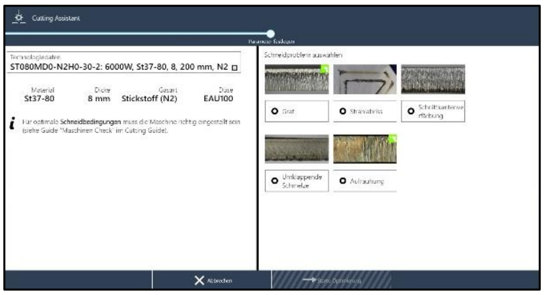
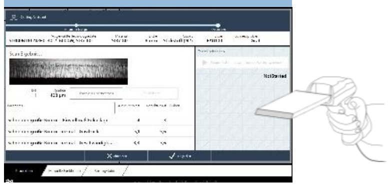
Easy handling due to dialog-guided cutting data optimization

Targeted adjustment of parameters based on the specific cutting issue and material





AI mode



Bandwidth mode

01 Start

Scan the cutting edge with the handheld scanner

02 Select cutting issue

- Select Laser Technology Table (LTT)
- **Select from a variety of cutting issues** such as burrs, roughness, or beam interruption

03 Iterative optimization process

- **AI mode:** Burr and roughness measurement and parameter recommendation by AI model
- **Bandwidth mode:** Cutting a series of test parts. Also suitable for cutting issues where the cutting edge cannot be scanned

Application spectrum

Sheet thickness in mm	1	2	3	4	5	6	8	10	12	15	20	25	30	40
AI mode					Highspeed and MD5									
Mild Steel Nitrogen	□	□	□	□	■	■	■	■	■	■	□	□	□	
Mild Steel Oxygen	□	□	□	□	□	□	□	□	□	□	□	□	□	□
Stainless Steel Nitrogen	□	□	□	□	□	□	□	□	□	□	□	□	□	□

Sheet thickness in mm	1	2	3	4	5	6	8	10	12	15	20	25	30	40
Bandwidth mode	Complete process range													
Mild Steel Nitrogen	■	■	■	■	■	■	■	■	■	■	■	■	■	
Mild Steel Mixed Gas					■	■	■	■	■	■	■	■		
Mild Steel Oxygen	■	■	■	■	■	■	■	■	■	■	■	■	■	■
Mild Steel Air					□	□	□	□	□	□	□	□		
Stainless Steel Nitrogen	■	■	■	■	■	■	■	■	■	■	■	■	■	■
Aluminum Nitrogen	□	□	□	□	□	□	□	□	□	□	□	□	□	□
Aluminum Gasmix			□	□	□	□	□	□	□	□	□	□		
Aluminum Oxigen	□	□	□	□	□	□	□							
Aluminum Air	□	□	□											

■ Available
□ Available from March 2026

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